GRAND CANYON NATIONAL PARK
WORLD HERITAGE SITE

REHABILITATE NORTH RIM INFRASTRUCTURE

VOLUME 2 – PIPELINES – NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)

VOLUME #	DESCRIPTION
VOLUME 0	GENERAL
VOLUME 1	PIPELINES - NORTH RIM
VOLUME 2	PIPELINES - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 3	PIPELINES - NORTH RIM TO SUPAI TUNNEL
VOLUME 4	PIPELINES - ROARING SPRINGS CAVE TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 5	ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 6	NORTH RIM WATER TREATMENT PLANT
VOLUME 7	NOT USED
VOLUME 8	NORTH RIM WASTEWATER COLLECTION SYSTEM
VOLUME 9	POWER DISTRIBUTION - NORTH RIM
VOLUME 10	POWER DISTRIBUTION - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 11	POWER DISTRIBUTION - ROARING SPRINGS PUMPHOUSE (RSPH) TO MANZANITA
VOLUME 12	POWER SYSTEM - COTTONWOOD CAMPGROUND

SHEET	SUB SHEET	SUB AREA	TITLE OF SHEET
1	020G001	GENERAL	SHEET INDEX
2	020D101	UTILITIES	RAW WATER HYDRAULIC PROFILE
3	020D102	UTILITIES	FINISHED WATER HYDRAULIC PROFILE
4	021C103	UTILITIES	BORE ALIGNENT OVERVIEW
5	021C104	UTILITIES	BORE PROFILE OVERVIEW
6	021C105	UTILITIES	BORE PROFILE OVERVIEW
7	021C401	UTILITIES	BORE ENTRY SITE LAYOUT
8	021C402	UTILITIES	BORE RECEIVING SITE LAYOUT AND EQUIPMENT STAGING PLAN
9	021C502	UTILITIES	WELLHEAD AND DRILLING CELLAR DETAIL
10	021C503	UTILITIES	TERMINUS AND CASING SPACER DETAILS

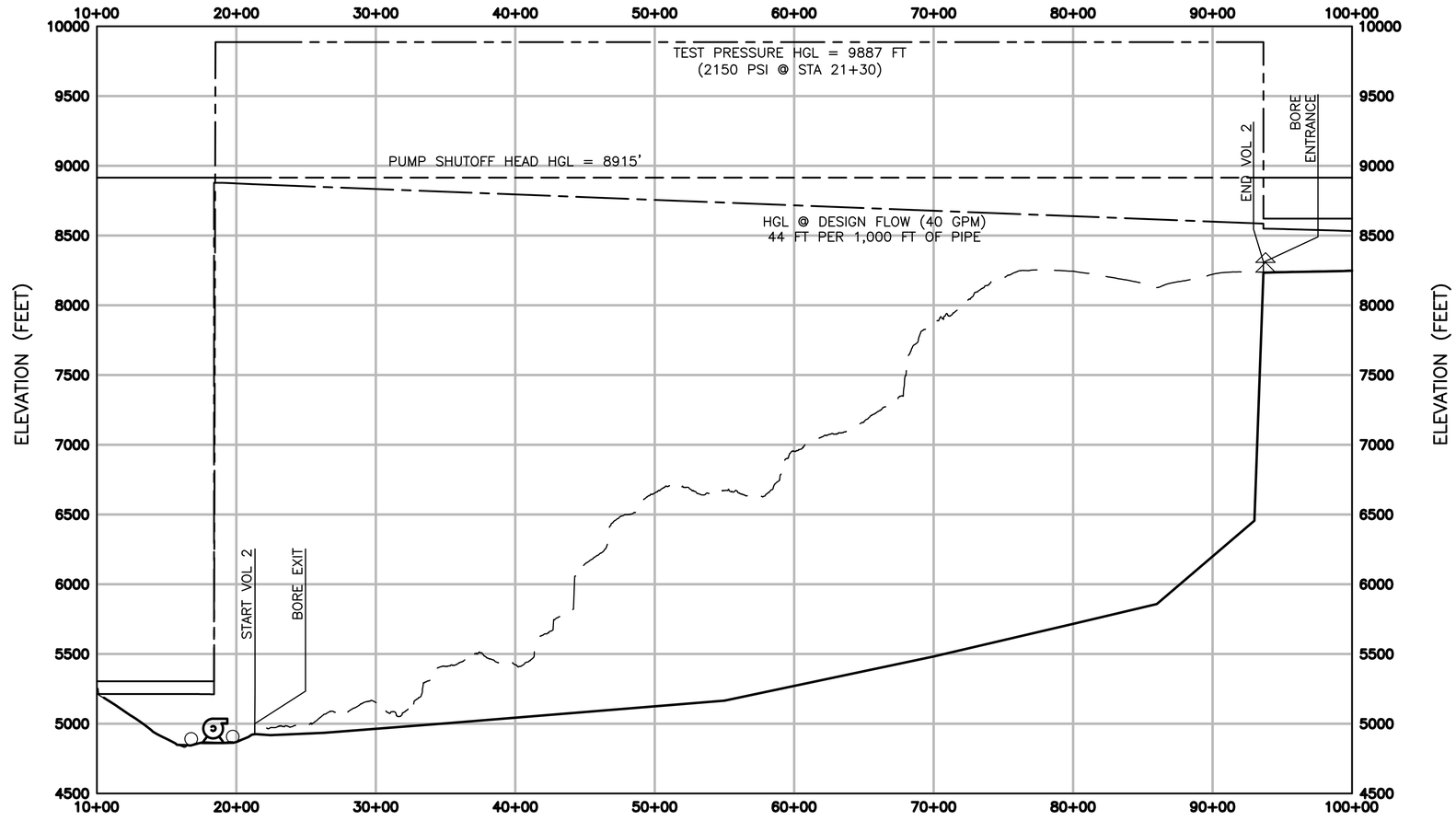
Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY Date Signed 8-1-2026 ARIZONA, U.S.A.

Nicholas Humphrey

Registered PROFESSIONAL ENGINEER (Civil) CERTIFICATE NO. 41809 RYAN RHOADES Date Signed 8-1-26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PRING CADD: N LYNCH TECH. REVIEW: M. GOUGH DATE: AUG 2026	SUB SHEET NO. 020 G001	TITLE OF SHEET GENERAL SHEET INDEX NORTH RIM - GRAND CANYON NATIONAL PARK
			DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 1 OF 10

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BORE ENTRY TO BORE EXIT HYDRAULIC PROFILE

LEGEND:

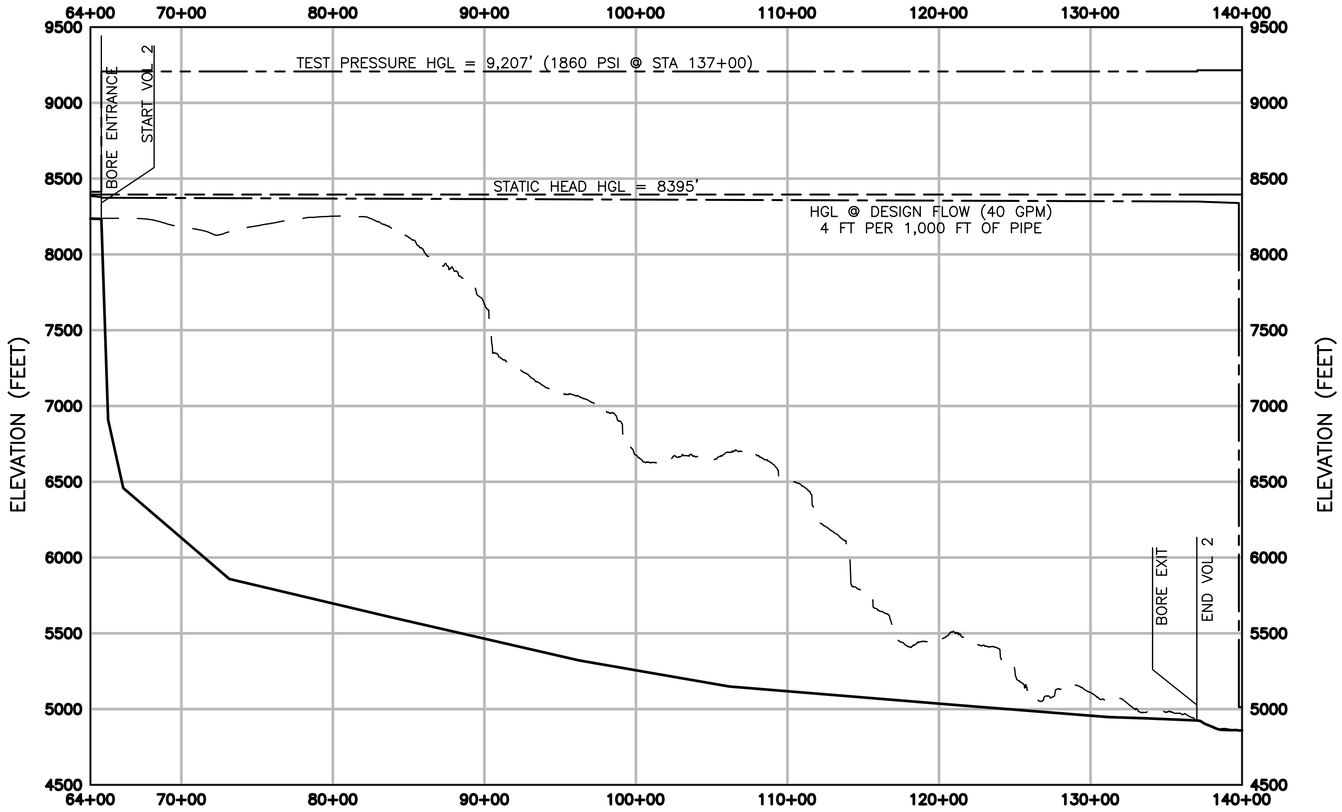
- PUMP
- COMBINATION AIR RELEASE AND VACUUM VALVE
- ISOLATION VALVE
- EXISTING GROUND AT SURFACE
- PIPE CENTERLINE
- WATERLINE HGL
- TEST PRESSURE
- PUMP SHUTOFF HEAD

500 0 500 1000 HORIZ.
500 0 500 1000 VERT.
SCALE OF FEET
H: 1"=500' V: 1"=500'

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 9-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUB: GEI CONSULTANTS GOLDEN, CO	DESIGNED: N. HUMPHREY CADD C. WITTENMEIER TECH. REVIEW: M. GOUGH/R. RHODES DATE: MAY 2026	SUB SHEET NO. 020 D101	TITLE OF SHEET GENERAL BORE B – RAW WATER HYDRAULIC PROFILE NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 2 OF 10

4/16/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\PROCESS\020-D101.dwg



BORE ENTRY TO BORE EXIT HYDRAULIC PROFILE

LEGEND:

- EXISTING GROUND AT SURFACE
- PIPE CENTERLINE
- WATERLINE HGL
- TEST PRESSURE
- STATIC HEAD

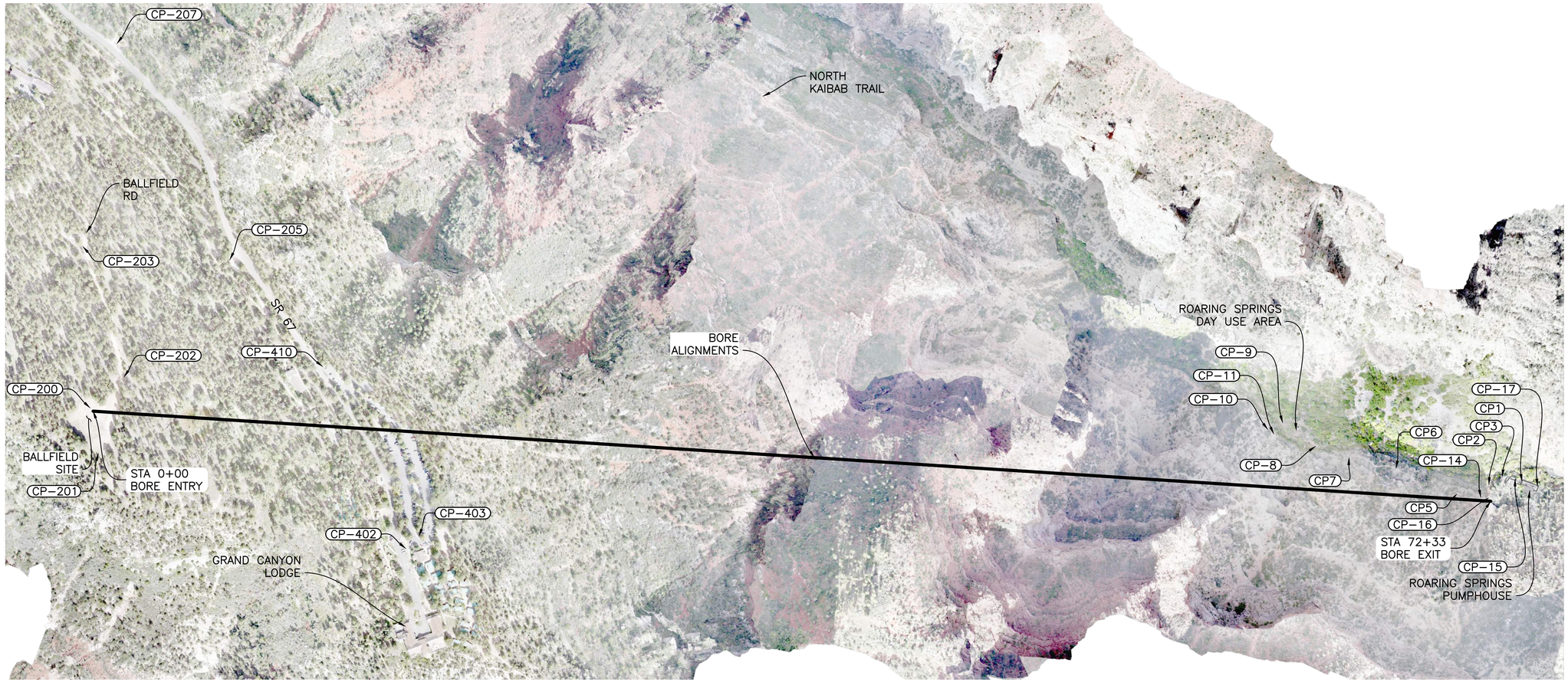
500 0 500 1000 HORIZ.
500 0 500 1000 VERT.
SCALE OF FEET
H: 1"=500' V: 1"=500'

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUB: GEI CONSULTANTS GOLDEN, CO	DESIGNED: N. HUMPHREY CADD C. WITTENMEIER TECH. REVIEW: M. GOUGH/R. RHODES DATE: MAY 2026	SUB SHEET NO. 020 D102	TITLE OF SHEET GENERAL BORE A – FINISHED WATER HYDRAULIC PROFILE NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 3 OF 10	

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GENERAL NOTES:

A. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED ON SHEETS 002C101 THROUGH 002C105 IN VOLUME 0.

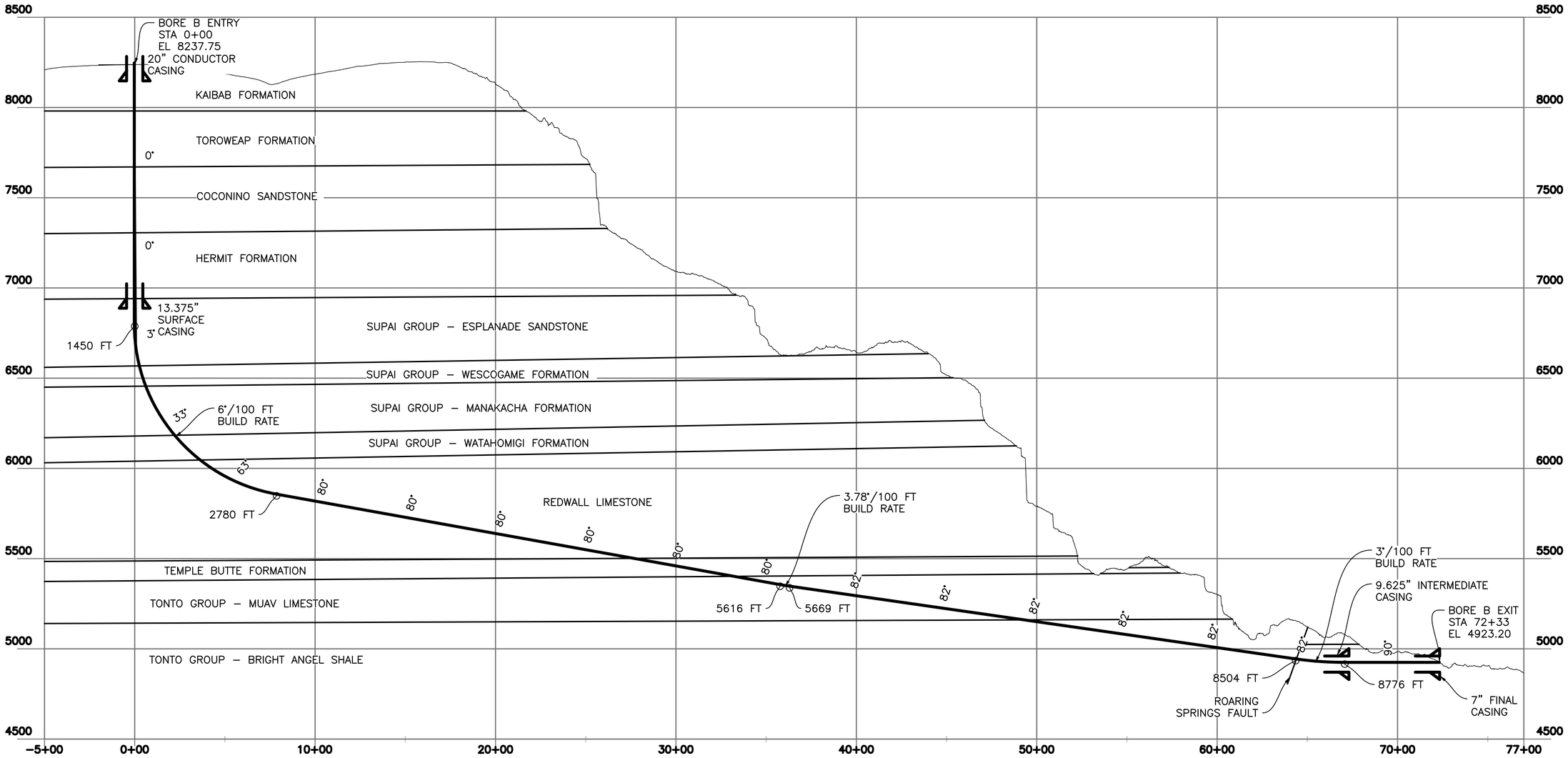
B. REFER TO VOLUMES 1 AND 5 FOR GESC PLAN ON SHEETS 011C102 AND 051C101.

300 0 300 600
SCALE OF FEET
1"=300'

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUB: GEI CONSULTANTS GOLDEN, CO	DESIGNED: A. PRING CADD C. WITTENMEIER TECH. REVIEW: P. BEARDEN DATE: MAY 2026	SUB SHEET NO. 021 C103	TITLE OF SHEET UTILITIES BORE ALIGNMENT OVERVIEW NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 4 OF 10

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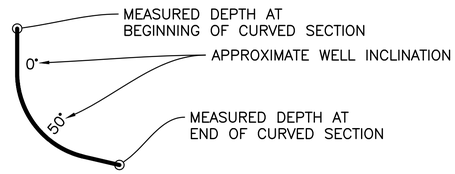


BORE B NOTES:

A. CASING DIAMETERS AND MEASURED DEPTH (TOTAL LENGTH).
   20 INCH CONDUCTOR CASING, 26 INCH BOREHOLE – 90 FT
   13.375 INCH SURFACE CASING, 17.5 INCH BOREHOLE – 1,350 FT
   9.625 INCH INTERMEDIATE CASING, 12.25 INCH BOREHOLE – 8,800 FT
   7 INCH FINAL CASING, 8.5 INCH BOREHOLE – 9,302 FT.

B. SEE GEOLOGIC INTERPRETATION SUMMARY REPORT BY GEI CONSULTANTS FOR BASELINE FORMATION DESCRIPTIONS, MECHANICAL PROPERTIES, GEOLOGIC CONDITIONS, AND GROUND BEHAVIORS.

SYMBOLS LEGEND:



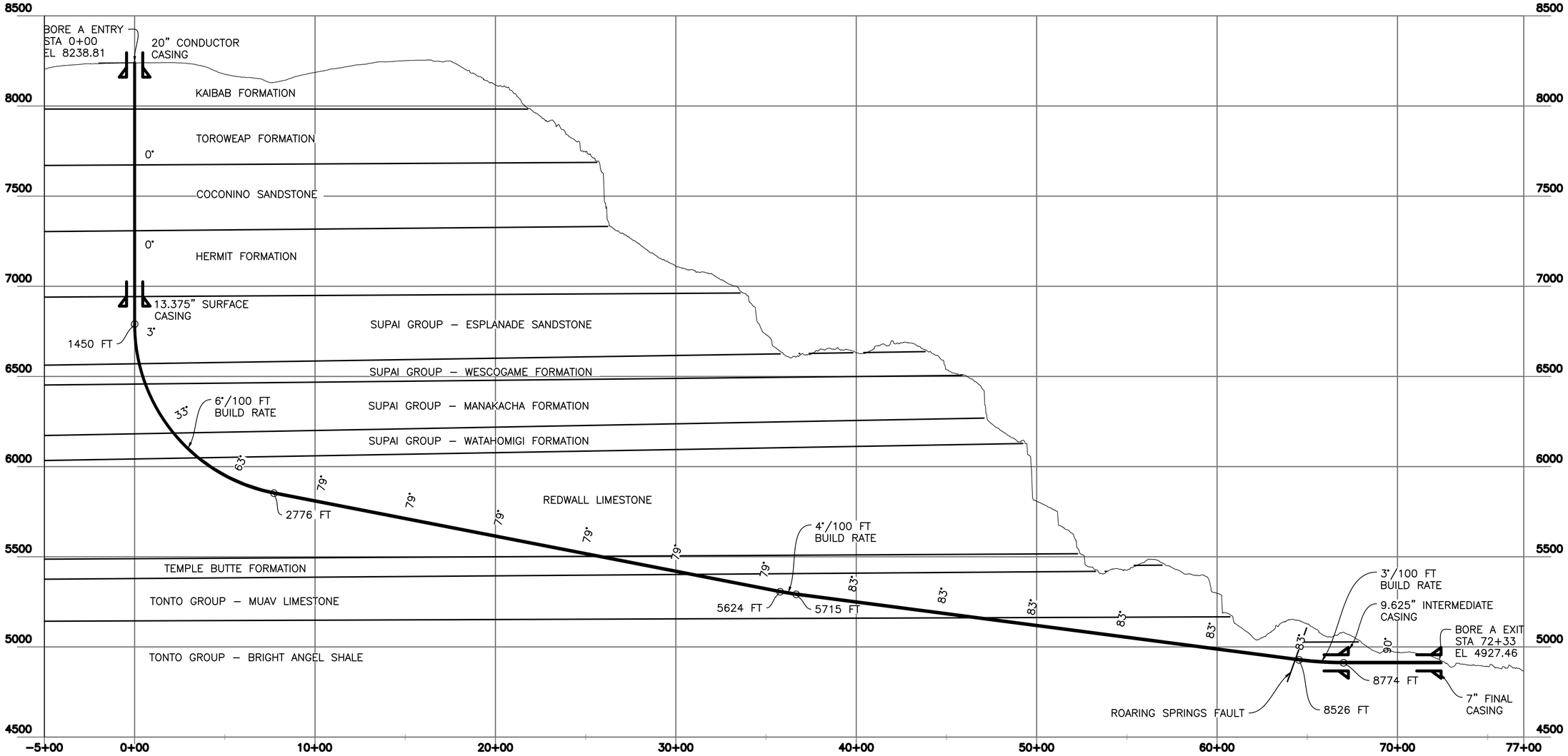
END OF CASING STRING

300 0 300 600
SCALE OF FEET
1" = 300'

Registered Professional Engineer's Certificate No. 56297 NATHAN SOULE Date Signed 06/30/26 ARIZONA U.S.A. Expires 06/30/26

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	A. PRING	021 C104	UTILITIES BORE B PROFILE OVERVIEW	113 182752
SUB: GEI CONSULTANTS GOLDEN, CO	C. WITTENMEIER			PMIS/PKG NO. 318719
	TECH. REVIEW: P. BEARDEN			SHEET 5 OF 10
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

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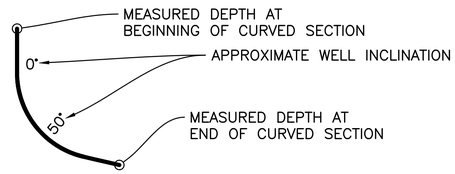


BORE A NOTES:

A. CASING DIAMETERS AND MEASURED DEPTH (TOTAL LENGTH).
   20 INCH CONDUCTOR CASING, 26 INCH BOREHOLE – 80 FT
   13.375 INCH SURFACE CASING, 17.5 INCH BOREHOLE – 1,350 FT
   9.625 INCH INTERMEDIATE CASING, 12.25 INCH BOREHOLE – 8,800 FT
   7 INCH FINAL CASING, 8.5 INCH BOREHOLE – 9,314 FT.

B. SEE GEOLOGIC INTERPRETATION SUMMARY REPORT BY GEI CONSULTANTS FOR BASELINE FORMATION DESCRIPTIONS, MECHANICAL PROPERTIES, GEOLOGIC CONDITIONS, AND GROUND BEHAVIORS.

SYMBOLS LEGEND:



END OF CASING STRING

300 0 300 600
SCALE OF FEET
1" = 300'

Registered Professional Engineer (Civil) CERTIFICATE NO. 56297 NATHAN SOULE Date Signed 06/30/26 ARIZONA U.S.A. Expires 06/30/2027

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	A. PRING	021 C105	UTILITIES BORE A PROFILE OVERVIEW	113 182752
SUB: GEI CONSULTANTS GOLDEN, CO	C. WITTENMEIER			PMIS/PKG NO. 318719
	TECH. REVIEW: P. BEARDEN			SHEET 6 OF 10
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

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GENERAL NOTES:

A. BOREHOLE B ENTRY LOCATION IS FIXED. BOREHOLE A SHALL BE LOCATED A MAXIMUM OF 40 FT FROM BOREHOLE B AND SHALL NOT BE LOCATED FURTHER SOUTH THAN BOREHOLE B.

B. CONTRACTOR TO SURVEY BOREHOLE A LOCATION PRIOR TO THE START OF DRILLING.

C. CONTRACTOR MAY CHOOSE TO DRILL BOREHOLE A OR B FIRST DEPENDING ON THEIR PREFERENCES.

KEYNOTES (X)

1. PRIOR TO CONSTRUCTION, CONTRACTOR SHALL COORDINATE WITH PARK TO ESTABLISH SENSITIVE RESOURCE AREA BOUNDARY AND INSTALL PARK–APPROVED TAPE TO BLOCK OFF AREA FROM CONSTRUCTION ACTIVITIES.

2. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE.

CONSTRUCTION LIMITS TABLE

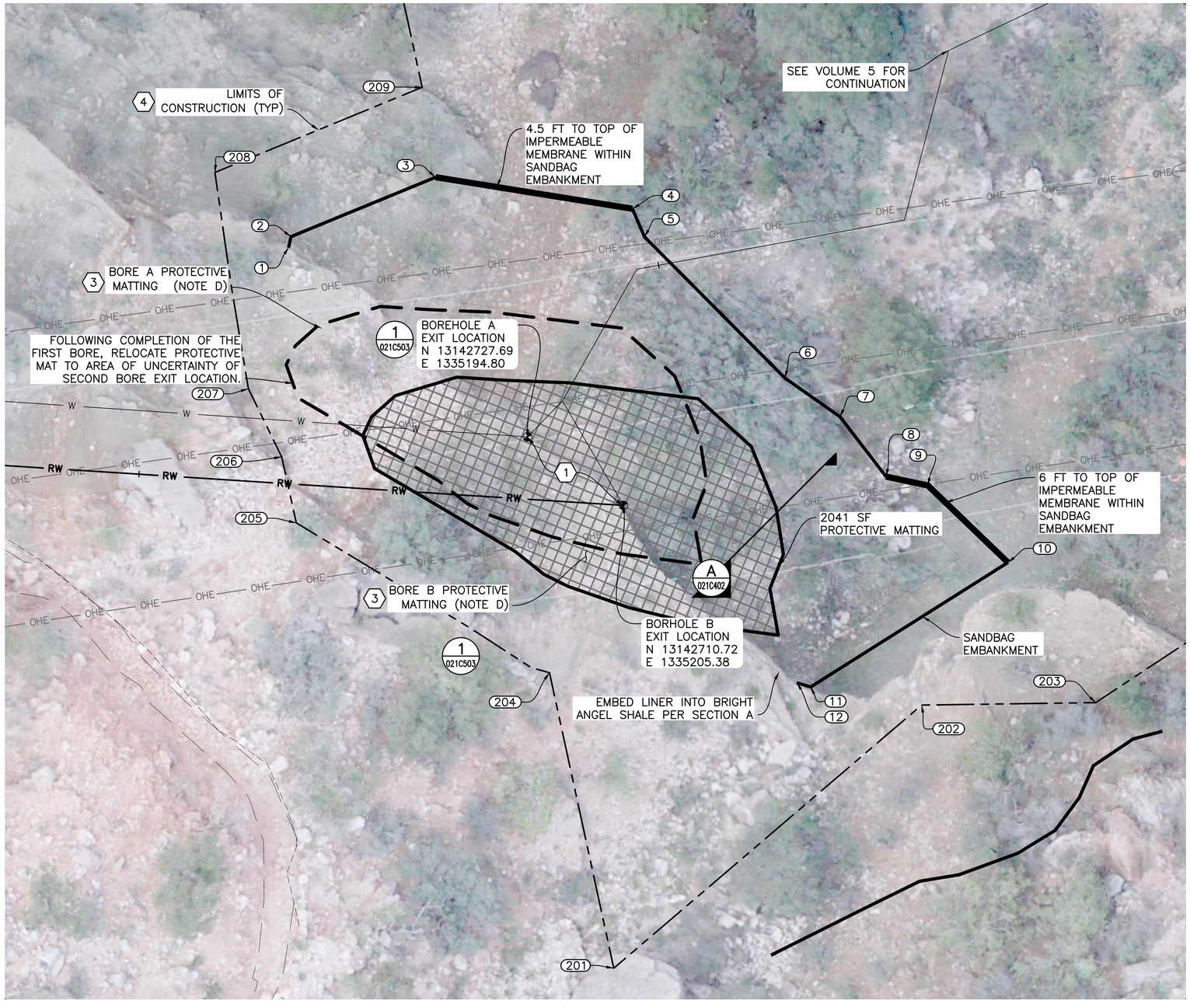
POINT NO.	NORTHING	EASTING
1	13146248.92	1328947.54
2	13146248.92	1328575.26
3	13145677.75	1328580.41
4	13145677.75	1328830.41
5	13145806.35	1328830.41
6	13145837.61	1328835.55
7	13145979.43	1328912.19
8	13146043.95	1328958.02
9	13146256.90	1328999.71

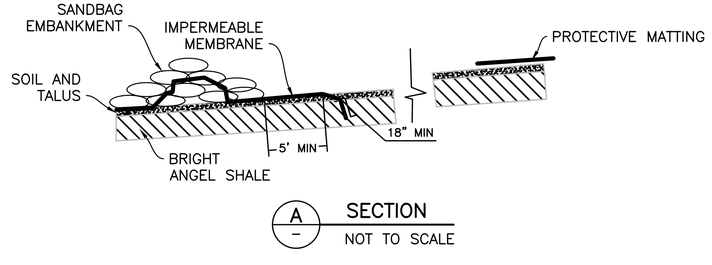
50 0 50 100
SCALE OF FEET
1"=50'

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUB: GEI CONSULTANTS GOLDEN, CO	DESIGNED: A. PRING CADD C. WITTENMEIER TECH. REVIEW: P. BEARDEN DATE: MAY 2026	SUB SHEET NO. 021 C401	TITLE OF SHEET UTILITIES BORE ENTRY SITE LAYOUT NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 7 OF 10

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A/– SECTION
NOT TO SCALE

PROTECTIVE MATTING AND SANDBAG EMBANKMENT POINT TABLE			
POINT NO.	NORTHING	EASTING	DESCRIPTION
1	13142772.96	1335169.64	SANDBAG EMBANKMENT
2	13142774.49	1335170.72	SANDBAG EMBANKMENT
3	13142774.40	1335197.28	SANDBAG EMBANKMENT
4	13142756.67	1335226.08	SANDBAG EMBANKMENT
5	13142751.48	1335226.15	SANDBAG EMBANKMENT
6	13142720.23	1335238.97	SANDBAG EMBANKMENT
7	13142710.56	1335245.11	SANDBAG EMBANKMENT
8	13142698.17	1335248.45	SANDBAG EMBANKMENT
9	13142694.05	1335254.43	SANDBAG EMBANKMENT
10	13142676.77	1335261.99	SANDBAG EMBANKMENT
11	13142670.06	1335222.95	SANDBAG EMBANKMENT
12	13142671.61	1335221.06	SANDBAG EMBANKMENT

KEYNOTES ⟨X⟩

1. SECOND BORE SHALL BE OFFSET NO MORE THAN 20 FEET FROM THE FIRST BORE EXIT (MEASURED PERPENDICULAR TO THE FIRST BORE ALIGNMENT).
2. NEITHER BORE SHALL EXIT WITHIN EXISTING DRAINAGES.
3. SIZE AND LOCATE BLAST MATTING TO EXTEND 5 FEET BEYOND THE AREA OF UNCERTAINTY OF THE FIRST BORE EXIT. AFTER FIRST BORE HAS BEEN COMPLETED, RELOCATE BLAST MATTING TO EXTEND 5 FEET BEYOND THE AREA OF UNCERTAINTY OF THE SECOND BORE EXIT.
4. AVAILABLE CONSTRUCTION LIMITS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE.

CONSTRUCTION LIMITS TABLE		
POINT NO.	NORTHING	EASTING
201	13142638.74	1335173.77
202	13142660.00	1335239.17
203	13142649.08	1335266.64
204	13142689.27	1335182.90
205	13142729.25	1335152.93
206	13142740.53	1335154.93
207	13142753.41	1335154.14
208	13142789.52	1335162.95
209	13142789.39	1335201.00
210	13142822.69	1335207.02

GENERAL NOTES.

A. LAYOUT OF EMBANKMENTS AND PROTECTIVE MATTING IS INDICATIVE. CONTRACTOR TO POSITION EMBANKMENTS IN THE FIELD TO PREVENT FLUID ESCAPING BORE EXIT FROM REACHING ROARING CREEK.

B. CONTRACTOR TO ONLY REMOVE AND DISPOSE OF TREES WITHIN THE LIMITS OF CONSTRUCTION. TREE REMOVAL TO BE KEPT TO A MINIMUM.

C. AREAS OF UNCERTAINTY REPRESENTED BY AREA WHERE A 20 FOOT RADIUS SPHERE INTERSECT THE GROUND SURFACE. CONTRACTOR MAY REDUCE RADIUS OF UNCERTAINTY BASED ON SELECTED SURVEY AND TRACKING METHODS AND WITH CONTRACTING OFFICE APPROVAL.

D. PROTECTIVE MATTING TO CONSIST OF BLAST MATTING OR APPROVED EQUIVALENT AND MEET MINIMUM REQUIREMENTS PER SPECIFICATION 31 70 00 – DIRECTIONAL DRILLING AND CARRIER PIPE INSTALLATION.

E. IMPERMEABLE MEMBRANE MUST HAVE MINIMUM PUNCTURE AND HYDRAULIC CONDUCTIVITY REQUIREMENTS PER SPECIFICATION 31 71 00 – DIRECTIONAL DRILLING AND CARRIER PIPE INSTALLATION.

F. CONTRACTOR TO TEMPORARILY STORE EQUIPMENT AND MATERIALS REQUIRED AT BORE EXIT LOCATIONS WITHIN THE LIMITS OF CONSTRUCTION AND IN A POSITION WHICH DOES NOT IMPEDE HELICOPTER ACCESS OR WORK PERFORMED BY OTHERS.

G. UNLESS OTHERWISE NOTED, IMPERMEABLE MEMBRANE WITHIN SANDBAG EMBANKMENT SHALL BE 3 FEET HIGH, MINIMUM.

H. IMPERMEABLE MEMBRANE SHALL BE EMBEDDED A MINIMUM OF OF FIVE FEET FROM UPSLOPE TOE OF SANDBAG EMBANKMENT AS SHOW IN DETAIL A.

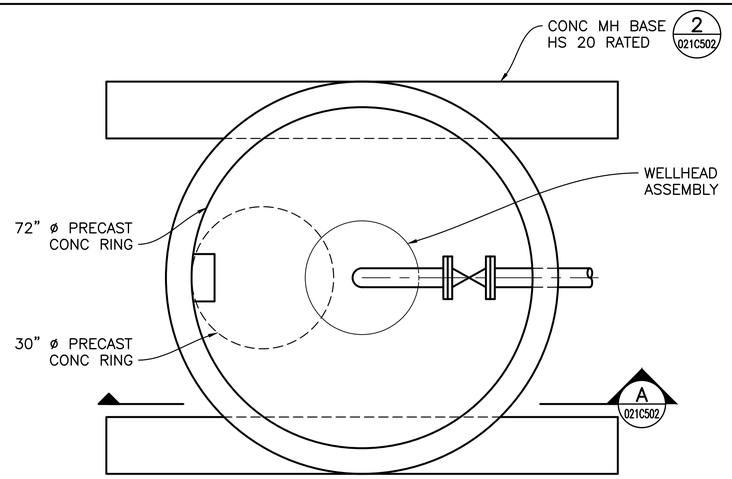
10 0 10 20
SCALE OF FEET
1"=10'

Registered Professional Engineer (Civil) CERTIFICATE NO. 55297 NATHAN SOULE Date Signed 05/30/2026 ARIZONA U.S.A. Expires 06/30/2027

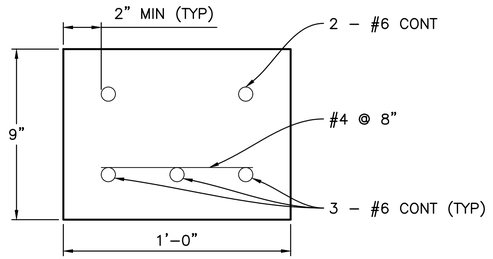
Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A. Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202 SUB: GEI CONSULTANTS GOLDEN, CO	DESIGNED: A. PRING CADD C. WITTENMEIER TECH. REVIEW: P. BEARDEN DATE: MAY 2026	SUB SHEET NO. 021 C402	TITLE OF SHEET UTILITIES BORE RECEIVING SITE LAYOUT AND EQUIPMENT STAGING PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 8 OF 10

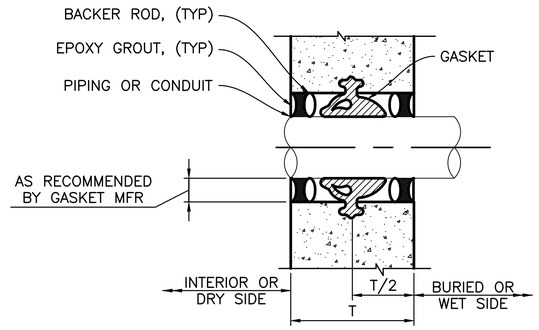
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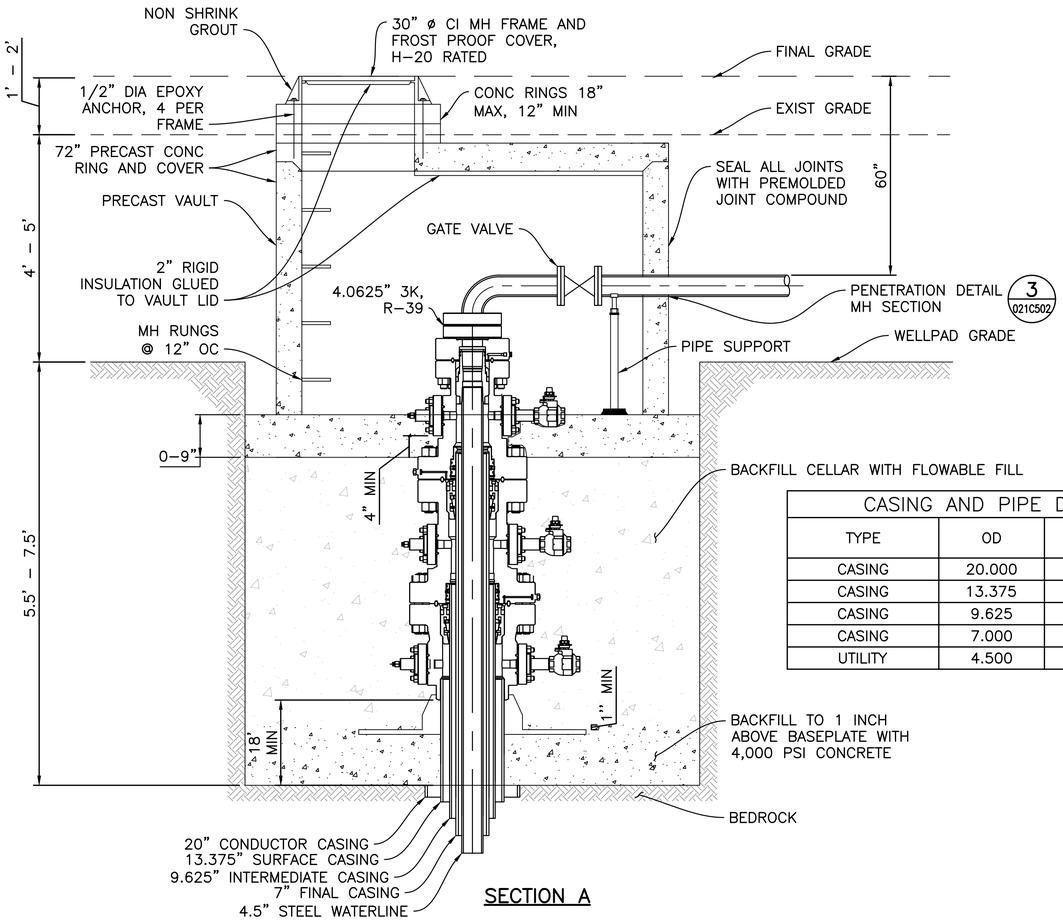
PLAN



2 021C502 FOOTING DETAIL
NOT TO SCALE



3 021C502 PENETRATION DETAIL MH SECTION
NOT TO SCALE



SECTION A

CASING AND PIPE DIMENSIONS			
TYPE	OD	GRADE	WEIGHT (LB/FT)
CASING	20.000	J–55	94.00
CASING	13.375	J–55	54.50
CASING	9.625	J–55	40.00
CASING	7.000	L–80	23.00
UTILITY	4.500	L–80	15.10

1 021C502 WELLHEAD AND DRILLING CELLAR
NOT TO SCALE

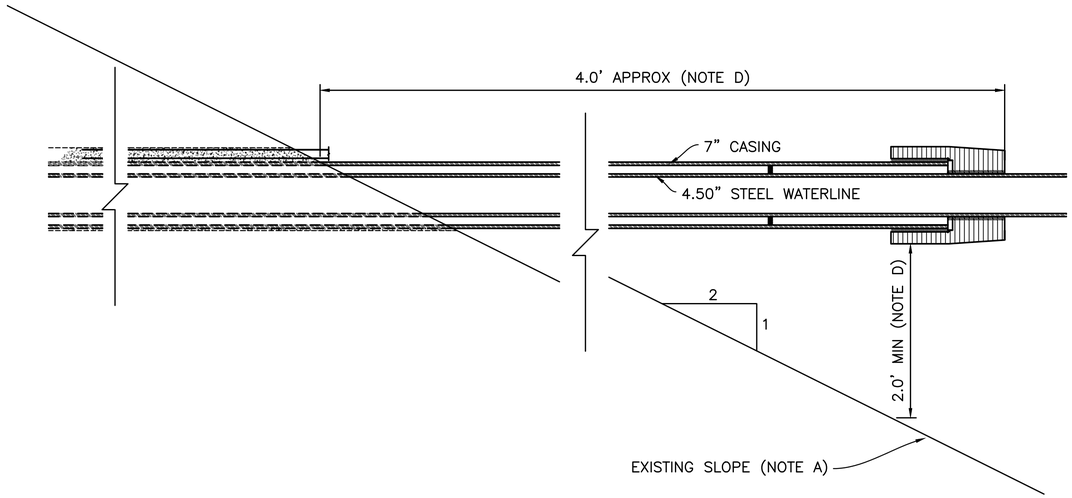
Registered Professional Engineer (Civil) CERTIFICATE NO. 65297 NATHAN SOULE Date Signed 05/30/2026 ARIZONA U.S.A. Expires 06/30/2027

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY Date Signed 5-1-2026 ARIZONA, U.S.A.

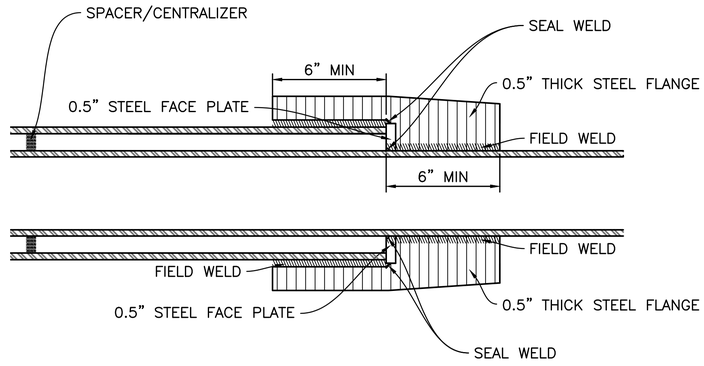
Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	A. PRING	021 C502	UTILITIES WELLHEAD AND DRILLING CELLAR DETAIL	113 182752
SUB: GEI CONSULTANTS GOLDEN, CO	CADD A. PRING			PMIS/PKG NO. 318719
	TECH. REVIEW: P. BEARDEN			SHEET 9 OF 10
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

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1 — BORE TERMINUS
NOT TO SCALE



2 — TERMINUS DETAIL
NOT TO SCALE

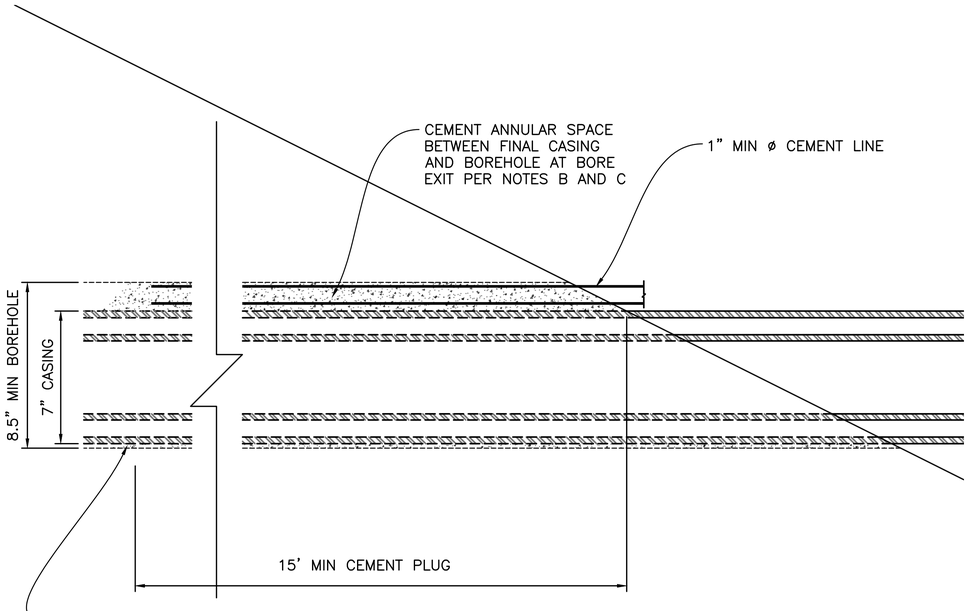
GENERAL NOTES:

A. GRADE OF SLOPE TO BE DETERMINED UPON BORE EXIT. 2:1 SLOPE SHOWN IS APPROXIMATELY THE SHALLOWEST GRADE WITHIN THE EXPECTED EXIT AREA.

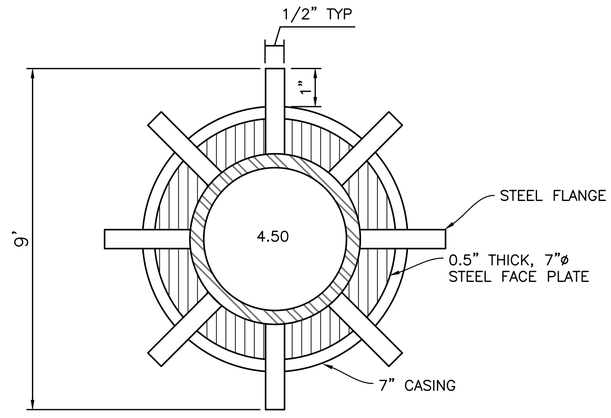
B. CEMENT FINAL CASING BACK TO 100 FEET BEYOND (ABOVE) INTERMEDIATE CASING SHOE, AT A MINIMUM.

C. CONTRACTOR SHALL CONSTRUCT 15 FOOT LONG MINIMUM CEMENT PLUG WITHIN ANNULUS BETWEEN BOREHOLE AND FINAL CASING PRIOR TO CEMENTING FROM BOREHOLE ENTRY LOCATION.

D. EXTEND CASING BEYOND BOREHOLE EXIT SUCH THAT 2 FEET MINIMUM CLEARANCE AND SUFFICIENT WORKING SPACE BELOW THE LOWEST POINT OF THE STEEL TERMINUS IS ACHIEVED.



3 — BOREHOLE EXIT GROUTING
NOT TO SCALE



4 — TERMINUS SECTION
NOT TO SCALE

CASING AND PIPE DIMENSIONS			
TYPE	MATERIAL	OD (IN)	ID (IN)
CASING	STEEL	7.000	6.366
UTILITY		4.500	3.826

Registered Professional Engineer (Civil) CERTIFICATE NO. 55297 NATHAN SOULE ARIZONA U.S.A. Date Signed 05/30/2026 Expires 06/30/2027

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	A. PRING	021 C503	UTILITIES TERMINUS AND CASING SPACER DETAILS	113 182752
SUB: GEI CONSULTANTS GOLDEN, CO	CADD: A. PRING		NORTH RIM - GRAND CANYON NATIONAL PARK	PMIS/PKG NO. 318719
	TECH. REVIEW: P. BEARDEN			SHEET 10 OF 10
	DATE: MAY 2026			

4/16/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\021-C502.dwg